fusionTrack UART interface - atracConnect

Document Information

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1	Initial version	LCO 2017-12-05	SSO 2017-12-11	NA
2	Added chapter 5 and 6 Parity bit corrected in chapter 3.1 UART protocol	LCO 2018-01-17	NA	MHA 2018-01-23
3	Correction of socket marking in chapter “Pinning of the atracConnect plug on the fusionTrack” and Illustration 10	LCO 2018-05-17	NA	MHA 2018-05-18
4	Changed Illustration 4. Added chapter 5.3 Sending exposure and trigger signals on EIO_1/EIO_2 lines	LCO 2018-07-24	NA	MHA 2018-08-06
5	Updated following modification applied on Application Firmware 0x01020008: - updated chapters 3.2.4, 4.1 and Table 3. - added chapters 3.2.5 and 4.2.	LBO 2023-06-01	LCO 2023-06-09	MLA 2024-07-03

mla Digitally signed by mla Date: 2024.07.03 15:01:14 +02'00'

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1 Purpose

The fusionTrack 500 is able to externally provide different signals via its EIO port. The signals can be the exposure of the camera; the strobe of the IR LEDs; a user trigger or the device timestamp. A UART communication allows to configure which signals are configured on the EIO lines.

To easily access these signals, it is recommended to use the atracConnect board whilst it is plugged to the fusionTrack 500 via an atracCable.

This document explains how the configuration of the EIO lines can be made via the UART interface and how the atracConnect board can be used to retrieve these signals.

2 Related products

Product	Manufacturer
fusionTrack 500	Atracsys
atracConnect	Atracsys
atracCable	Atracsys

Table 1: related products

3 UART interface

3.1 UART protocol

The settings of the UART communication are the following:

- Speed: 2 Mbaud (default, can be changed over a setting in the fusionTrack)
- Transmission length: 8 bits
- Stop bit length: 1 bit
- Parity check: Odd
- Bits 0-7 are inverted. To transmit a '1', the line is low ('0') and to transmit a '0', the line is high ('1').

Start bit	Bit 0	Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Parity bit	Stop bit
0	Data				Command				Odd	1

Table 2: UART frame

3.2 UART commands

The commands are sent to the fusionTrack via the EIO_3 line (see chapter 4 atracConnect Board)

Command Name	Command bits 7..4	Data bits 3..0	Description	Default value
Enable	0000	0000	Disabled	Disabled
		xxx1	Enabled	
Exposure Mode	0001	0000	Disabled	Disabled
		0001	Enabled	
Continuous Mode	0010	0000	Disabled	Disabled
		xxx1	Delay (data×10us)	
Signal select on EIO_1 for transmission	0011	0000	Timestamp exposure	Timestamp exposure
		0001	Timestamp continuous	
		0010	Strobe	
		0011	Trigger	
		0100	Exposure	
		0101	PPS	
Signal select on EIO_2 for transmission	0100	0000	Timestamp exposure	Timestamp continuous
		0001	Timestamp continuous	
		0010	Strobe	
		0011	Trigger	
		0100	Exposure	
		0101	PPS	
Signal select on EIO_1 for reception	0101	0000	TAG single mode	Disabled
		0001	TAG dual mode	
		1111	Port disable	
Signal select on EIO_2 for reception	0110	0000	TAG single mode	Disabled
		0001	TAG dual mode	
		1111	Port disable	

Table 3: UART commands

3.2.1 Command "Enable"

This command enables or disables the transmission of timestamp/signals by the fusionTrack.

3.2.2 Command "Exposure mode"

This command enables or disables the transmission of the timestamp exposure.

Refer to chapter 3.3 UART timestamp transmission for more information about the timestamp transmission.

3.2.3 Command "Continuous mode"

This command enables or disables the transmission of the continuous timestamp. The delay between two transmissions of timestamps can be specified. The allowed delays are between 10-150us.

Refer to chapter 3.3 UART timestamp transmission for more information about the timestamp transmission.

3.2.4 Command "Signal select on EIO_x for transmission"

This command specifies which signals are available on the lines. The possibilities are:

- **Timestamp exposure**. The timestamp is transmitted synchronously with the exposure of the camera.
- **Timestamp continuous**. The timestamp is transmitted continuously with the specified delay.
- **Strobe**. The internal strobe signal is available on the line. Internal to the fusionTrack, the strobe signal drives the IR LEDs.
- **Trigger.** A user settable trigger is available on the line. (cf Chapter 5.2)
- **Exposure.** The internal camera exposure signal is available on the line.
- **PPS.** The Pulse Per Second signal generated by the PTP module when this one is enable.

Note:

- Using this command automatically set the port in transmission mode.

3.2.5 Command "Signal select on EIO_x for reception"

This command specifies how the signals available on the lines in reception mode should be interpreted.

The possibilities are:

- **TAG single mode**: The selected port enter the TAG single mode interpretation meaning the incoming signal on this port is interpreted as triggers. The arrival times of each trigger are tagged meaning the local timestamp of each trigger rising edge is stored and transmitted further with a corresponding ID. In this “single” mode, both port can handle different trigger signals (both trigger signals are considered unrelated).
- **TAG dual mode**: The selected port enter the TAG dual mode interpretation. For this mode, the triggers are stored in a small dual ring buffer. This mode allow to run the triggers faster (the double of the rate allowed in the single mode) without losing any information.
- **Port disable:** The selected port is disabled. The incoming signals are received but not processed (this can be seen as an open drain input).

Note:

- Using this command automatically set the port in reception mode.

3.3 UART timestamp transmission

The fusionTrack provides a 64-bits timestamp. It counts from the device start up at 0 with a granularity of 1 microsecond.

The least significant byte (LSB) is sent first.

Start bit	Bits 0-7	Parity bit	Stop bit
0	Byte 0	odd	1

Start bit	Bits 0-7	Parity bit	Stop bit
0	Byte 1	odd	1

...

Start bit	Bits 0-7	Parity bit	Stop bit
0	Byte 7	odd	1

4 atracConnect Board

Both EIO_1 and EIO_2 ports can be used for transmission and reception modes independently. EIO_3 port is always reserved for UART communication. In the next two sections are presented the possible connection for using the transmission and reception modes.

4.1 EIO_1 and EIO_2 in transmission mode (EIO_3 for commands)

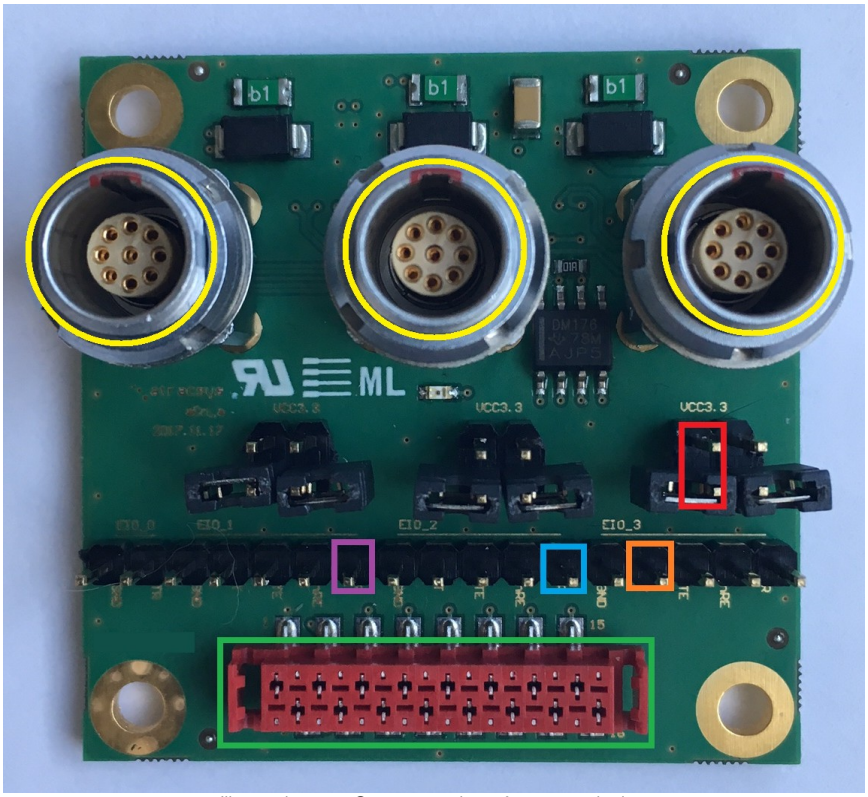


Illustration 1: aCn connections for transmission

To use the atracConnect board in transmission for the fusionTrack UART interface, please follow this instruction:

1. Plug a jumper on the EIO_3 header 2×2 between TE and VCC3.3 (framed in red on Illustration 1).
2. Plug the atracCable on one of the three LEMO connectors (circled in yellow on Illustration 1).
3. The UART commands are sent via EIO_3_T (framed in orange on Illustration 1).
4. EIO_1 signal is available on EIO_1_R (framed in purple on Illustration 1).
5. EIO_2 signal is available on EIO 2_R (framed in blue on Illustration 1).

Note that all the signals are also available on the micomatch connector (framed in green on Illustration 1). The pining of the connector is available in chapter 6 Simplified schematics of the atracConnect board.

4.2 EIO_1 and EIO_2 in reception mode (EIO_3 for commands)

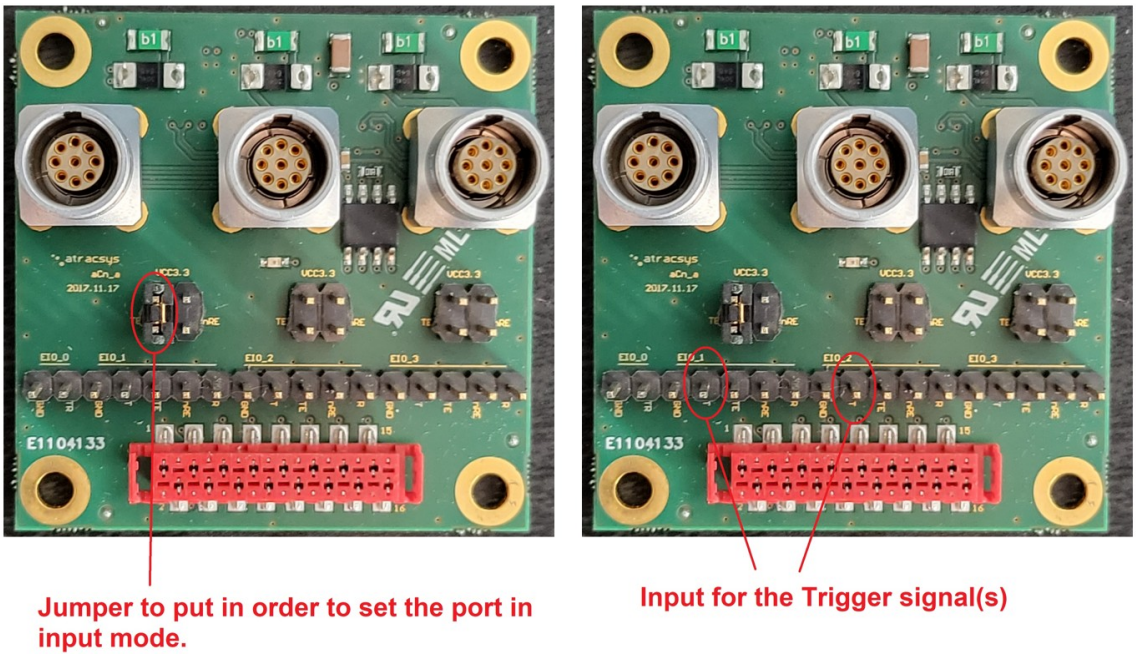


Illustration 2: aCn connections for reception

To use the atracConnect board in reception for the fusionTrack UART interface, please follow this instruction:

1. Plug a jumper on the EIO_3 header 2×2 between TE and VCC3.3 (framed in red on Illustration 1).
2. Plug the atracCable on one of the three LEMO connectors (circled in yellow on Illustration 1).
3. The UART commands are sent via EIO_3_T (framed in orange on Illustration 1).
4. For setting EIO_1 in reception, plug a jumper on the EIO_1 header 2×2 between TE and VCC3.3 (see Illustration 2 left picture)
5. For setting EIO_2 in reception, plug a jumper on the EIO_2 header 2×2 between TE and VCC3.3 (the same way it is done for EIO_1)
6. Wires transmitting external signals (i.e. triggers for the tagging reception mode) can be connected to the EIO_1_T or EIO_2_T pin depending which one is set in reception mode (see Illustration 2 right picture)

Note:

When using the dual tagging mode on one port, the other can not used either the single or dual tagging mode as all the internal electronic is reserved for the other port.

5 Example of use with RealTerm console

This chapter quickly describes how to communicate with the fusionTrack using the RealTerm console via the atracConnect board.

To transmit data from the computer to the fusionTrack, a USB to UART bridge is used. Usually these USB to UART bridges are not able to communicate at 2Mbaud (the default fTk UART speed). Therefore, the baudrate is set to a lower speed in this example .

The RealTerm console is available here: https://sourceforge.net/projects/realterm/.

5.1 Connections and configuration of UART speed

1. Start the fusionTrack and wait until it's ready to be used (When the device status led lights in solid green).

2. Connect the atracConnect to the fusionTrack with an atracCable. The atracConnect is then connected to the computer via the USB to UART bridge.

   On Illustration 3, the atracCable is connected to LEMO 1; UART TX (green cable) is connected to EIO_3_T; UART RX (yellow) is connected to EIO_1_R. A jumper on EIO_3 2×2 header connects TE to VCC3.3.

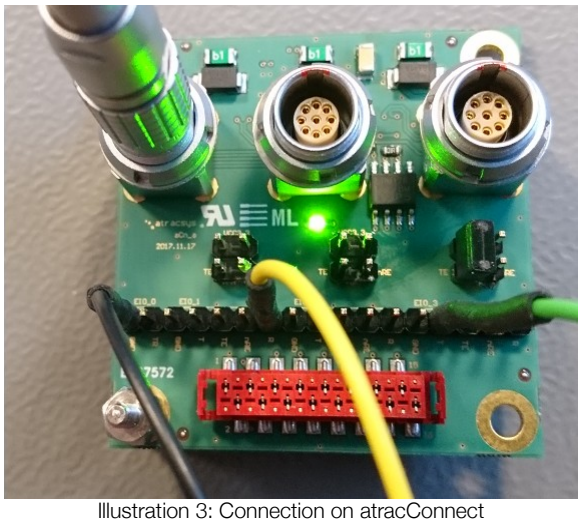


Illustration 3: Connection on atracConnect

3. Set the UART speed of the fusionTrack using the GUI demo. In the device tab of the option panel set the aCn UART speed to 57600 (b/s).

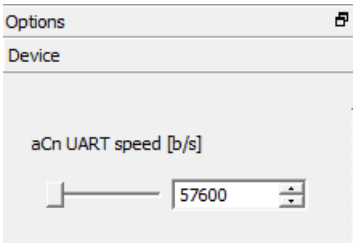


Illustration 4: Change the Baud Rate of the fusionTrack

4. In the "Display" tab of RealTerm, set the parameters visible in Illustration 5.

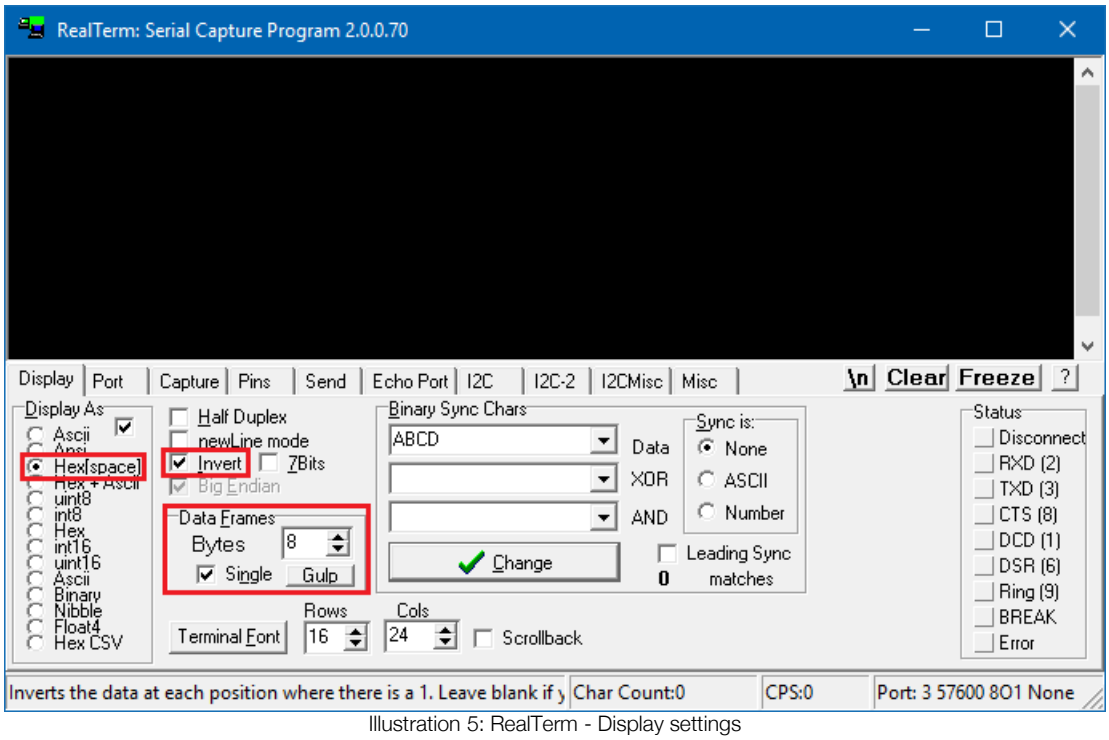


Illustration 5: RealTerm - Display settings

5. In the "Port" tab of the RealTerm, set the parameters visible in Illustration 6 and selects the port corresponding to the USB to UART bridge. Do not forget to click on the "Change" button.

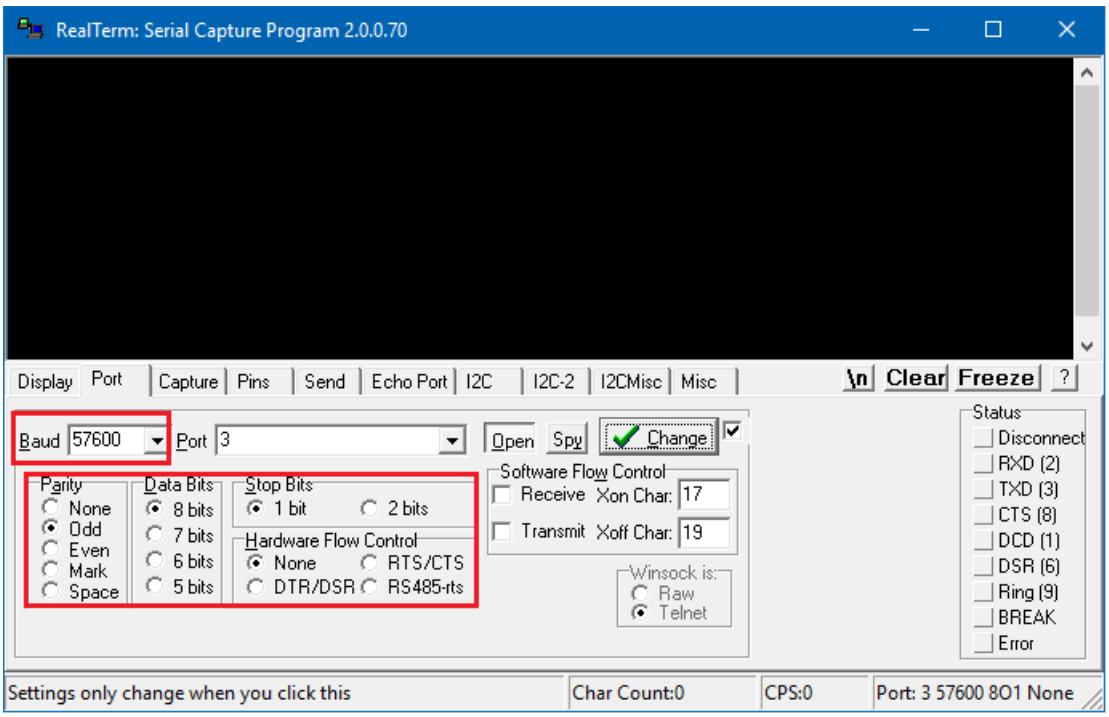


Illustration 6: RealTerm - Port settings

5.2 Sending timestamps on EIO_1/EIO_2 lines

In this example, the fusionTrack will be configured to send timestamps on its EIO port.

1. Follow all the steps described in chapter 5.1 Connections and configuration of UART speed.
2. Send the following commands by pressing "Send Numbers" in the "Send" tab of RealTerm.

Command	Description
254	General Enable
238	Enable Exposure Mode (transmitted by default on EIO_1)
208	Enable Continuous Mode (transmitted by default on EIO_2)

Note: As explained in chapter 3.1, the bits of the commands shown in Table 3 are inverted.

3. The timestamps are now printed in the RealTerm console

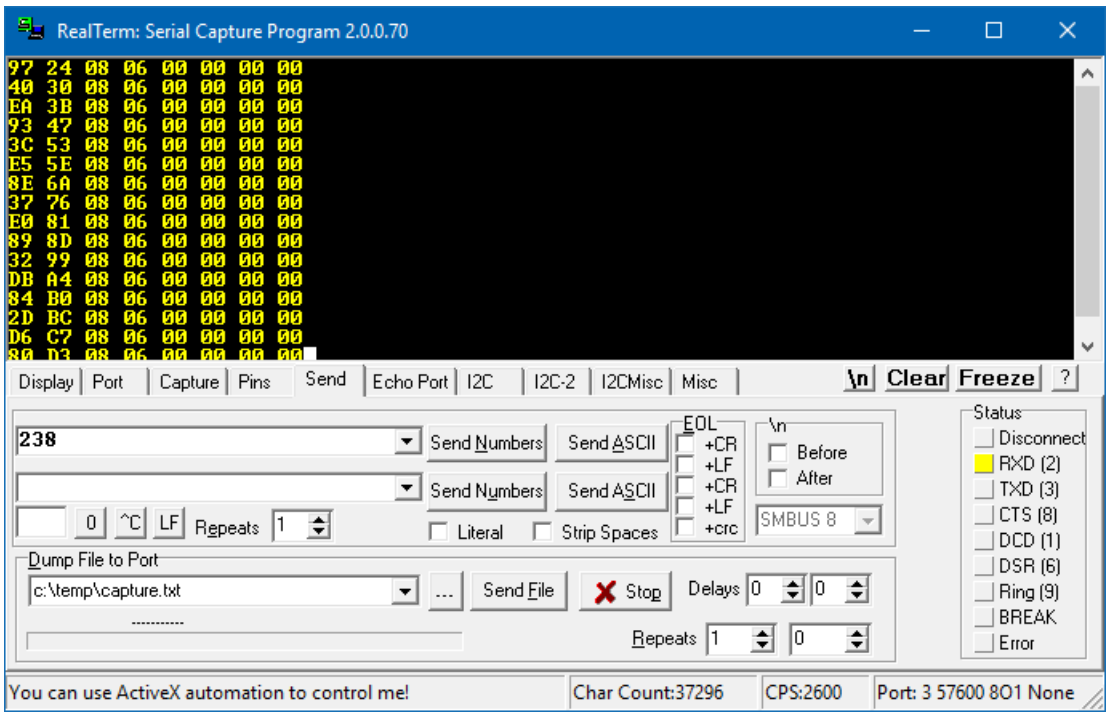


Illustration 7: RealTerm prints data received on EIO_1 line

Note: On Illustration 7 the timestamps are printed with the least significant byte first.

5.3 Sending exposure and trigger signals on EIO_1/EIO_2 lines

The fusionTrack can be configured to send a trigger to other equipment. The trigger is transmitted for each frame and can be configured using the fusionTrack GUI demo. The user can define the start time inside a frame and the duration of the trigger with a granularity of 1 microsecond.

This example shows how to configure the fusionTrack to send exposure and trigger signal on EIO port.

1. Follow all the steps described in chapter 5.1 Connections and configuration of UART speed.

2. Use the GUI demo to configure the trigger signal. The option aCn trigger start define the delay between the begining of the frame and when the trigger is set to '1'. The option aCn trigger duration define how long the signal stays high.

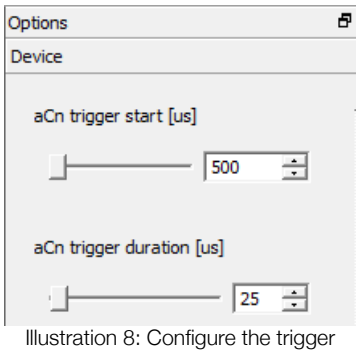


Illustration 8: Configure the trigger

3. Send the following commands by pressing "Send Numbers" in the "Send" tab of RealTerm.

Command	Description
254	General Enable
203	Send Exposure signal on EIO_1 line
188	Send Trigger signal on EIO_2 line

4. The exposure and the trigger is available on EIO_1 and EIO_2 lines.

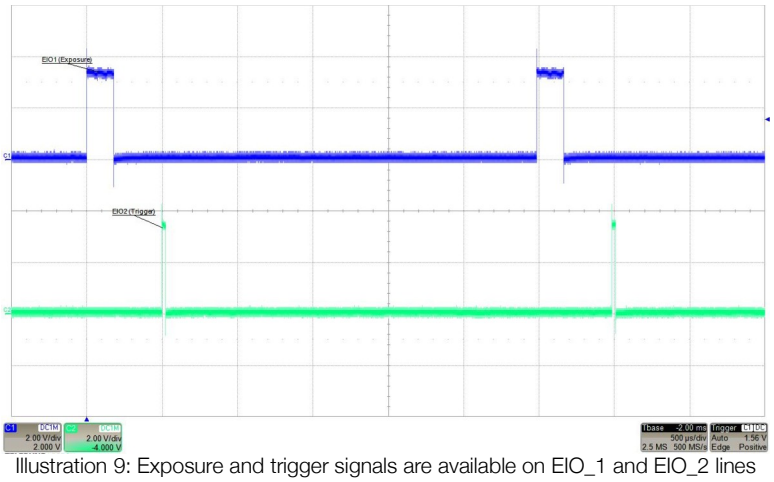
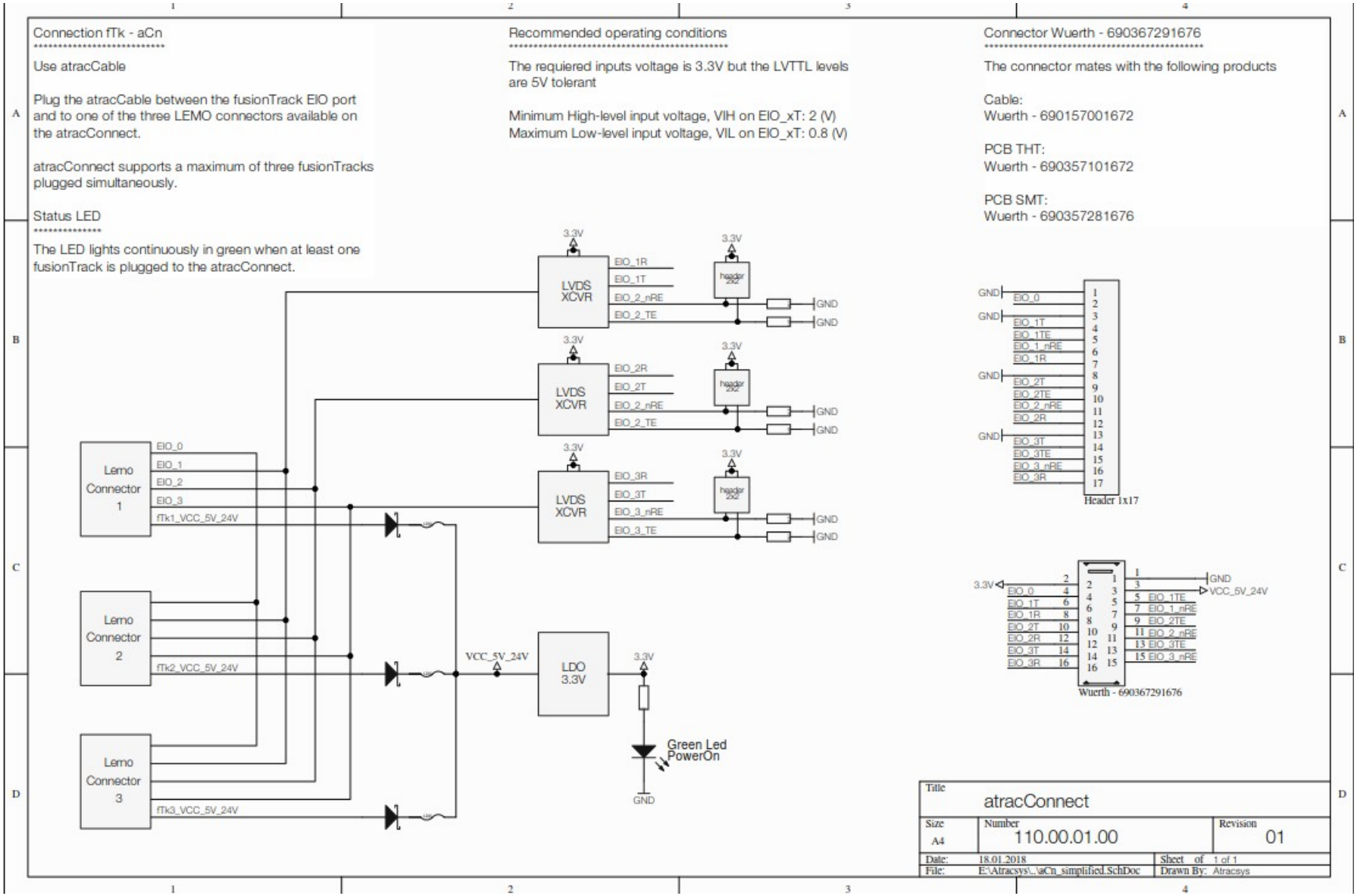


Illustration 9: Exposure and trigger signals are available on EIO_1 and EIO_2 lines

6 Simplified schematics of the atracConnect board



7 Pinning of the atracConnect plug on the fusionTrack

Pins	Name
1	EIO_2_P
2	EIO_2_N
3	EIO_3_N
4	EIO_3_P
5	VCC_5V / VCC_24V
6	EIO_0
7	EIO_1_N
8	EIO_1_P
9	GND

Illustration 10: Pinning (as seen when looking at the backside of the fusionTrack)

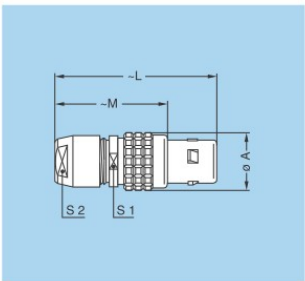
Illustration 10 shows the pinning of the atracConnect plug located on the backside of the fusionTrack. EIO_0 is an open-drain 3.3V connection. EIO_1, EIO_2, EIO_3 are three half-duplex bidirectional M-LVDS lines with a maximum data rate of 250Mbps per line. VCC_5V / VCC_24V and GND provides power from the fusionTrack to the atracConnect board.

A valid mating plug for this socket is the Lemo – FGG.0B.309.CLAD56 (https://www.lemo.com/pdf/FGG.0B.309.CLAD56.pdf).

Alternative plug are:

Short version

JGG Straight plug, short version, key (G), cable collet

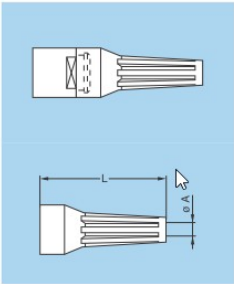


Reference		Dimensions (mm)				
Model	Series	A	L	M	S1	S2
JGG	0B	9.5	32	22	8	7

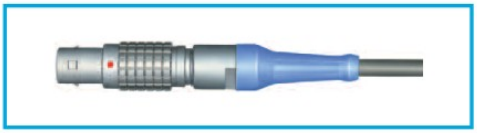
M4 Cable assembly (page 193)

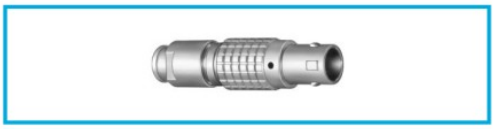
Connector with bend relief in polyurethane elastomer or in silicone

GM• Bend relief (TPU)

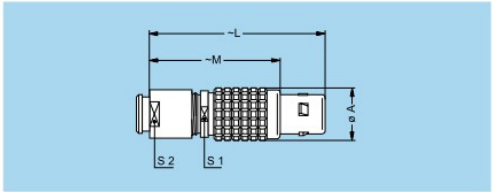


A bend relief made from thermoplastic polyurethane elastomer can be fitted over LEMO plugs and sockets that are supplied with nut for fitting such bend relief.
They are available in nine different colours match with the GRA insulating washers (see page 171).
Use the part numbers shown below to order this accessory separately.





FGG Straight plug, key (G) or keys (A…M), cable collet and nut for fitting a bend relief



Reference		Dimensions (mm)				
Model	Series	A	L	M	S1	S2
FGG	00[1)]	6.4	27.5	18.5	5.5	5
FGG	0B	9.5	35.0	25.0	8.0	7
FGG	1B	12.0	42.0	33.0	10.0	9
FGG	2B	15.0	48.0	36.0	13.0	12
FGG	3B	18.0	56.5	41.5	15.0	15
FGG	4B	25.0	71.0	53.0	21.0	20

Note: 1) the surface design of the 00 series is different.
The bend relief must be ordered separately (see page 175).

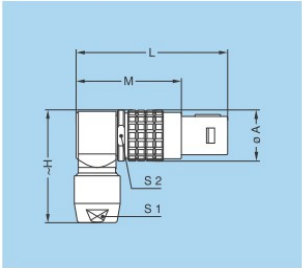
Ref.	Colour
A	blue
B	white
G	grey

Ref.	Colour
J	yellow
M	brown
N	black

Ref.	Colour
R	red
S	orange
V	green

90° variant
(one of different)

FSG Anglissimo right angle plug, key (G) or keys (A…M), cable collet



Reference		Dimensions (mm)					
Model	Series	A	H	L	M	S1	S2
FSG	00	8.0	18.1	24.8	16.8	5	7
FSG	0B	10.0	22.4	30.3	20.3	7	9
FSG	1B	12.0	26.4	36.5	25.5	9	11
FSG	2B	16.5	34.5	44.0	32.0	12	15

M5 Cable assembly (page 192)

More information see https://www.lemo.com/catalog/ROW/UK_English/unipole_multipole.pdf